Ferromagnetic scar tower under bulk dephasing: notes on the results so far

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Abstract

We study how the ferromagnetic quantum many-body scar tower of a spin-1/2 chain decays under uniform σ^z dephasing, using exact density-matrix evolution and full Liouvillian diagonalization for $L \leq 8$. The coherent oscillations launched from the scar state $|-\rangle^{\otimes L}$ decay at a rate that grows roughly linearly with L ($\Gamma \approx 0.8\,\gamma L$ in a fixed fit window), whereas non-scar probes (energy autocorrelation and a staggered transverse magnetization) decay at rates close to the single-spin value 2γ with weak or no L-dependence. At fixed $L = 6$ all rates are linear in γ for weak dephasing; for strong dephasing the scar rate falls back to 2γ. Several exact results organize the data: the two fully polarized states are dark states, so the steady state is three-fold degenerate, and the edge coherences of the tower are exact Liouvillian eigenmodes with $\mathrm{Re}\,\lambda = -2n\gamma$ for every L.

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1 Model

Hamiltonian. Spin-1/2 ring of L sites (periodic boundaries), $S_j^\alpha = \sigma_j^\alpha/2$, bond singlet projector $R_{j,j+1} = \frac{1}{4} - \mathbf{S}_j \cdot \mathbf{S}_{j+1}$:

$$H = H_I + hS^z_{\mathrm{tot}}, \qquad H_I = \sum_j \Big[\sum_\alpha \big(a_\alpha R_{j,j+1}\sigma^\alpha_{j+2} + b_\alpha \sigma^\alpha_{j-1} R_{j,j+1}\big) + \sum_{\alpha\beta} c^{(j)}_{\alpha\beta}\, \sigma^\beta_{j-1} R_{j,j+1} \sigma^\alpha_{j+2} \Big]. \quad (1)$$

Every term of H_I annihilates the ferromagnetic multiplet $|\Phi_n\rangle \propto (S^-_{\mathrm{tot}})^n |\uparrow\cdots\uparrow\rangle$, $n = 0, \ldots, L$, so $H|\Phi_n\rangle = h(L/2 - n)|\Phi_n\rangle$: an equally spaced tower of $L + 1$ exact scars. Coefficients: $h = 0.5$, $a = (0.73, 0.41, 0.29)$, $b = (0.58, -0.37, 0.66)$; the only nonzero c are c_{xz}, c_{zy}, c_{yx} with baseline values $0.31, -0.44, 0.27$.

Random c. To remove translation symmetry, each nonzero $c_{\alpha\beta}$ is multiplied on every bond j by an independent factor uniform in $[0.5, 1.5]$ (signs and scale kept; a, b uniform). One realization per L (seed $1000 \cdot 1 + L$), shared by all runs below unless stated otherwise. Every term still annihilates the tower.

Dephasing.

$$\dot{\rho} = \mathcal{L}\rho = -i[H, \rho] + \gamma \sum_j \big(\sigma_j^z \rho\, \sigma_j^z - \rho\big). \tag{2}$$

In the Pauli basis $\rho = 2^{-L} \sum_P r_P P$ the dissipator is diagonal, $\mathcal{D}(P) = -2\gamma\, n_{xy}(P)\, P$, with $n_{xy}(P)$ the number of sites carrying σ^x or σ^y. A single spin's transverse coherence decays as $e^{-2\gamma t}$.

Initial states. The scar state $|-\rangle^{\otimes L}$ (all spins along $-x$), which lies entirely in the tower, and product states on a cone around $-x$: spin j along $(-\cos\theta, \sin\theta\cos\varphi_j, \sin\theta\sin\varphi_j)$ with random φ_j; $\theta = 0$ is the scar state.

Methods. (i) Exact evolution of the real Pauli-coefficient vector, $\exp(\mathcal{L}\, dt)$ by a Taylor series summed to round-off; translation orbits for the uniform model (dim $\approx 4^L/L$), the full Pauli basis (4^L) for random c. (ii) Full exact diagonalization of the dense $4^L \times 4^L$ real generator (LAPACK `dgeev`, Intel MKL) for $L \le 6$; at $\gamma = 0$ the Liouvillian is degenerate, so H is diagonalized instead and the modes are $|m\rangle\langle n|$, $\lambda = -i(E_m - E_n)$.

2 Exact results

Dark states and steady states. $|\Phi_0\rangle = |\uparrow \cdots \uparrow\rangle$ and $|\Phi_L\rangle = |\downarrow \cdots \downarrow\rangle$ are eigenstates of H and of every σ_j^z, hence dark states of the dissipator. The kernel of $\mathcal{L}$ is spanned by

$$\mathbb{1}, \qquad |\Phi_0\rangle\langle\Phi_0|, \qquad |\Phi_L\rangle\langle\Phi_L|, \tag{3}$$

and ED finds exactly three zero eigenvalues for $L = 4, 5, 6$. The infinite-temperature state is therefore *not* the unique steady state. The coherence $|\Phi_0\rangle\langle\Phi_L|$ is an eigenmode with $\lambda = -2\gamma L - ihL$, which saturates the general bound $-2\gamma L \le \mathrm{Re}\,\lambda \le 0$.

Edge coherences of the tower. Since $\sigma_j^z|\Phi_0\rangle = |\Phi_0\rangle$ and $\sum_j \sigma_j^z = 2S^z_{\mathrm{tot}}$,

$$\mathcal{D}\big(|\Phi_0\rangle\langle\Phi_n|\big) = \gamma\, |\Phi_0\rangle\langle\Phi_n| \Big(\sum_j \sigma_j^z - L \Big) = -2n\gamma\, |\Phi_0\rangle\langle\Phi_n|, \tag{4}$$

and both states are eigenstates of H, so $|\Phi_0\rangle\langle\Phi_n|$ (and likewise $|\Phi_L\rangle\langle\Phi_{L-n}|$) is an exact Liouvillian eigenmode with

$$\lambda = -2n\gamma - ihn \qquad \text{for every } L. \tag{5}$$

The slowest tower modes therefore decay at exactly 2γ, independent of L; ED confirms four modes at $-\mathrm{Re}\,\lambda = 2\gamma$ for each L. In first-order perturbation theory the coherences $|\Phi_n\rangle\langle\Phi_{n+1}|$ decay at $\gamma L(1 - m_n m_{n+1})$ with $m_n = 1 - 2n/L$: 2γ at the edges, $\approx \gamma L$ in the middle of the tower. The scar signal is a binomially weighted sum of these, which makes it multi-exponential (Sec. 4).

Initial loss of tower weight. For $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = |-\rangle^{\otimes L}$, $w = \mathrm{Tr}(P_{\mathrm{tower}}\rho)$ and $[P_{\mathrm{tower}}, H] = 0$,

$$\dot{w}(0) = \gamma \sum_j \big(\|P_{\mathrm{tower}}\sigma_j^z|\psi\rangle\|^2 - 1 \big), \qquad P_{\mathrm{tower}}\sigma_j^z|\psi\rangle = \tfrac{2}{L} S^z_{\mathrm{tot}}|\psi\rangle \;\Rightarrow\; \dot{w}(0) = -\gamma(L-1), \tag{6}$$

using permutation symmetry of the tower and $\langle (S^z_{\mathrm{tot}})^2 \rangle = L/4$. Reproduced exactly by the numerics for $L = 4, \ldots, 8$. At late times $w \to (L+1)/2^L$.

Overlaps of the probes. $P_{\text{tower}}\sigma_j^x P_{\text{tower}} = \frac{2}{L} S^x_{\text{tot}} P_{\text{tower}}$ for every j, so $O_k = \sum_j e^{ikj}\sigma_j^x$ has zero tower part for $k \neq 0$. The single-site σ^x content of H is $\frac{1}{4}(a_x + b_x)\sum_j \sigma_j^x$ (the c terms contain none), so $\mathrm{Tr}(O_k H) = 0$ exactly for $k \neq 0$ even with random c. Since $P_{\text{tower}} H P_{\text{tower}} = h S^z_{\text{tot}} P_{\text{tower}}$ and $[P_{\text{tower}}, H] = 0$, the operator $\tilde{H} = H - h S^z_{\text{tot}} P_{\text{tower}} = (1 - P_{\text{tower}}) H (1 - P_{\text{tower}})$ is conserved at $\gamma = 0$ and has no tower component.

S^z is not conserved, but does not relax to the tower value. At $\gamma = 0$, $\langle \sigma^z \rangle$ from a cone state relaxes only partially (for $L = 5$, $\theta = \pi/2$: from 0.279 to a late-time mean ≈ 0.15), while its tower part is only $\frac{2}{L}\,\mathrm{Tr}(P_{\text{tower}} S^z_{\text{tot}} \rho_0) = 0.031$. The difference is the thermal value of the bulk: H contains $h S^z_{\text{tot}}$, so energy conservation fixes a nonzero S^z.

3 Scar oscillations and their decay

Noiseless check. At $\gamma = 0$ the scar state precesses perfectly, $\langle \sigma_j^x \rangle = -\cos ht$, $\langle \sigma_j^y \rangle = -\sin ht$, on every site and for every L (deviation $\lesssim 10^{-13}$), with uniform or random c.

Decay against L. The amplitude $m_\perp = \sqrt{m_x^2 + m_y^2}$ decays faster for larger L (Fig. 1). For the translation-invariant model, fitting $e^{-\Gamma t}$ while $m_\perp > 0.05$ gives the rates in Table 1, well described by $\Gamma \approx 0.63\,\gamma L^{1.13}$. Breaking translation with random c changes them by less than 2%.

L	4	5	6	7	8	9	10
Γ/γ, uniform c	3.06	3.80	4.66	5.75	6.68	7.61	8.39
Γ/γ, random c	3.06	3.89	4.68	5.81	6.69	–	–

Table 1: Scar decay rate at $\gamma = 0.01$, fit window $m_\perp > 0.05$.

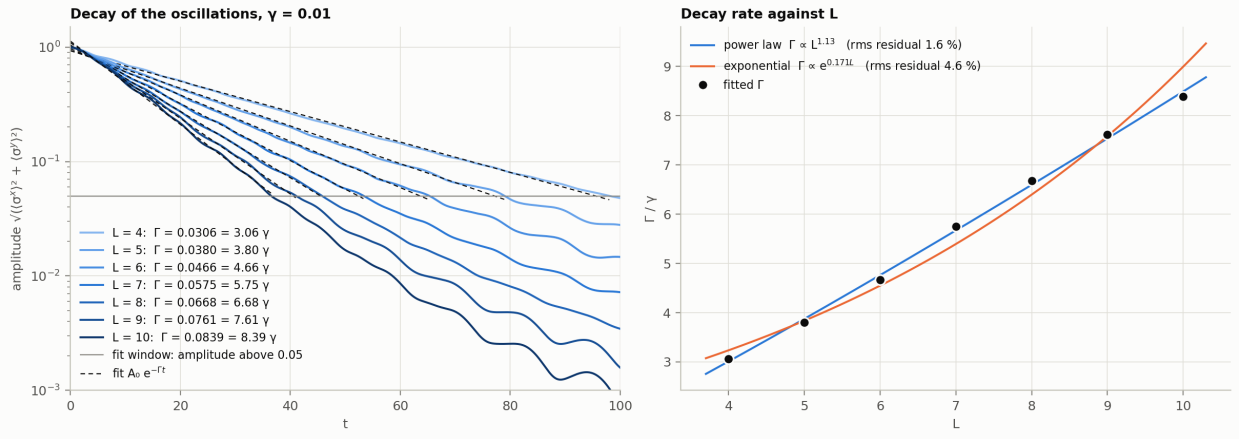


Figure 1: Uniform model, $\gamma = 0.01$: decay of $m_\perp$ from $|-\rangle^{\otimes L}$ for $L = 4, \ldots, 10$ and the fitted rate against L (`plot_scripts/fit_decay.py`).

Cone states. Starting off the tower (Fig. 2), the oscillations of $\langle \sigma^x \rangle$ survive at $\gamma = 0$ with reduced amplitude for small tilt ($\theta = \pi/8, \pi/4$) and are essentially gone for $\theta \geq 3\pi/8$, $L \geq 5$; with dephasing they die within a few periods. At small L the site averages carry a sampling offset from the random azimuths, $\sin\theta\,\overline{\cos\varphi_j}$ and $\sin\theta\,\overline{\sin\varphi_j}$.

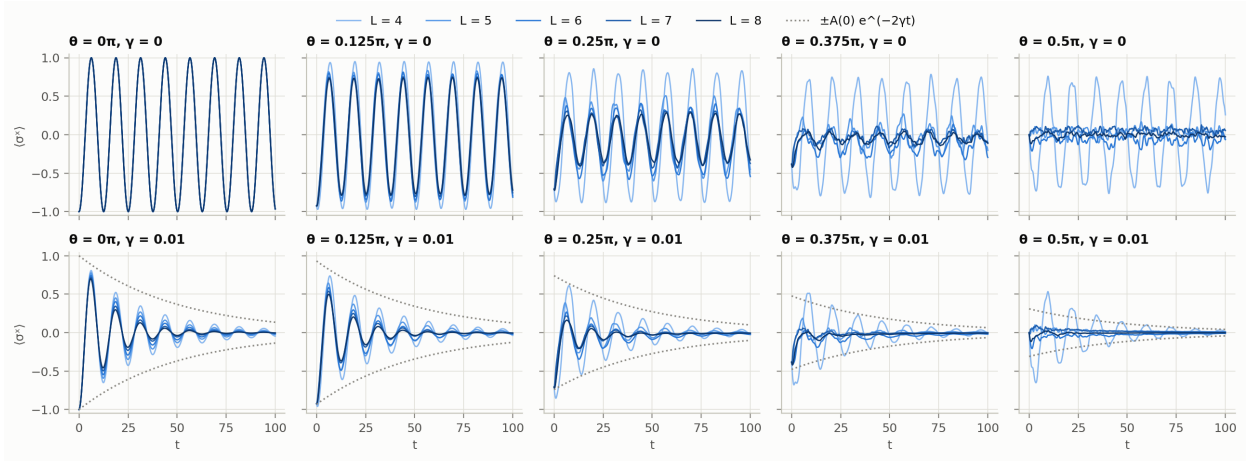


Figure 2: Random c: site-averaged $\langle\sigma^x\rangle$ from cone states with random azimuths, $\gamma = 0$ (top) and 0.01 (bottom); dotted: free-spin envelope (`plot_scripts/plot_disorder.py`).

4 Scar versus non-scar probes

Three probes, evolved with the same Lindbladian (random c, $L = 4, \ldots, 8$, $\gamma = 0$ and 0.01):

- **P1, scar:** $m_\perp(t)$ from $\rho_0 = |-\rangle\langle-|^{\otimes L}$.
- **P2, energy:** $C(t) = \mathrm{Tr}[\tilde{H} e^{\mathcal{L}^\dagger t}(\tilde{H})]/\,\mathrm{Tr}\,\tilde{H}^2$, conserved at $\gamma = 0$.
- **P3, thermal:** $|C^+(t)|$ for $O^+ = \sum_j (-1)^j \sigma_j^+$ ($k = \pi$, even L), built from separate evolutions of O_π^x and O_π^y so that the precession in h drops out. At $\gamma = 0$ it thermalizes by $t \approx 1$ and then sits at a finite-size floor ($0.17, 0.05, 0.015$ for $L = 4, 6, 8$). Its dephasing rate is $\Delta\Gamma$, the slope of $-\ln(|C_\gamma^+|/|C_0^+|)$ on a common short window.

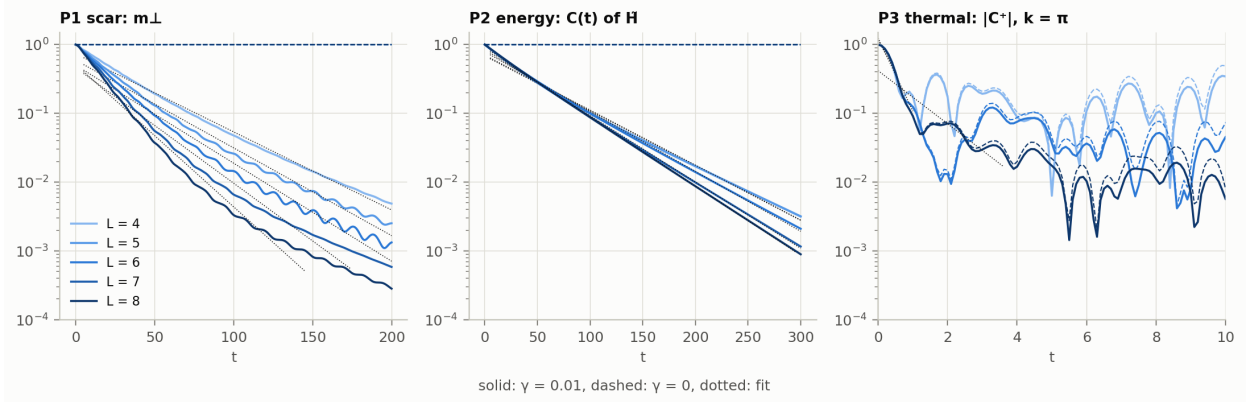


Figure 3: The three probes against time, $\gamma = 0.01$ (solid) and $\gamma = 0$ (dashed); dotted lines are the fits (`plot_scripts/fit_probes.py`).

Scaling with L. Table 2 and Fig. 4. The scar rate grows close to linearly in L; the non-scar rates stay near the single-spin value 2γ with a weak (energy) or no (thermal) L-dependence.

Caveats.

- The scar decay is multi-exponential (Fig. 3, left): mid-tower coherences ($\sim \gamma L$) dominate early, the exact 2γ edge modes late. Γ_{scar} and its L-exponent depend on the fit window.

probe (Γ/γ)	$L=4$	5	6	7	8	power law
P1 scar, window $t \in [5, 60]$	3.22	4.01	4.69	5.67	6.51	$L^{1.01\pm0.03}$
P1 scar, window $t \in [5, m_\perp = 10^{-3}]$	2.61	2.94	3.28	3.87	4.77	$L^{0.84\pm0.12}$
P2 energy $\tilde{H}$	1.83	1.84	2.00	2.22	2.32	$L^{0.37\pm0.07}$
P3 thermal $\Delta\Gamma$, $t \in [0, 0.5]$	1.43	–	1.59	–	1.61	$L^{0.18\pm0.06}$
P3 thermal $\Delta\Gamma$, $t \in [0, 1]$	2.25	–	2.25	–	1.85	$L^{-0.27\pm0.19}$

Table 2: Decay rates at $\gamma = 0.01$ (random c) and power-law fits $\Gamma = aL^b$.

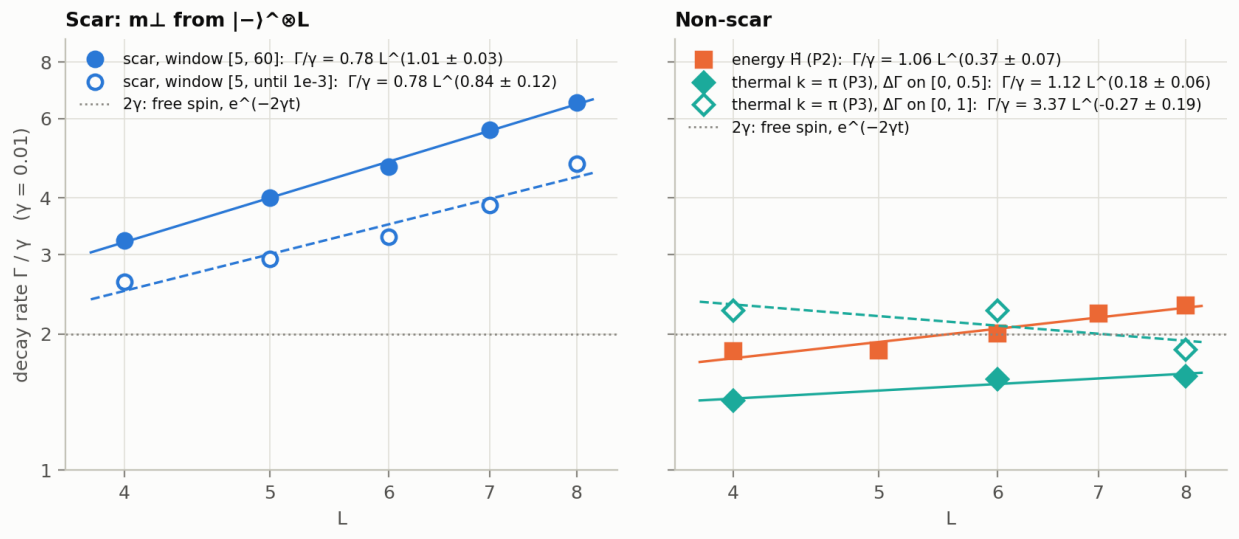


Figure 4: Decay rate against L (log-log) with power-law fits; scar (left) and non-scar probes (right).

- $L = 4, \ldots, 8$ is a factor of two in L: power law and exponential fit about equally well; the exponents are descriptive.
- One disorder realization per L.
- For P3, a window that ends at twice the noise floor lengthens with L and produces a spurious growth of $\Delta\Gamma$ $(1.43, 2.25, 4.39)$; only common windows are meaningful.
- The predicted initial rate of P2, $2\gamma \sum_P n_{xy} h_P^2 / \sum_P h_P^2$, is 2.7–2.9 γ; the long-time decay is slower (1.8–2.3 γ).

5 Liouvillian spectrum

Full ED at $\gamma = 0.01$ for $L = 4, 5, 6$ (Fig. 5). Each right eigenvector R is split into tower ($\|P_{\text{tower}} R P_{\text{tower}}\|$), mixed and bulk weights. Findings:

- three zero modes (Sec. 2), and $\mathrm{Re}\,\lambda \geq -2\gamma L$ saturated by $|\Phi_0\rangle\langle\Phi_L|$;
- the bulk is centred at $-\mathrm{Re}\,\lambda \approx \gamma L$ (the mean dephasing rate of a Pauli string);
- the slowest tower modes sit at exactly 2γ (edge coherences); the next ones at 4γ for $L = 5, 6$ (2.94γ at $L = 4$);
- the slowest bulk mode is at 1.67γ, 1.66γ, 1.88γ for $L = 4, 5, 6$, comparable to the P2 rate;
- modes with tower weight > 0.9: 22 of 25, 32 of 36, 38 of 49: at $\gamma = 0.01$ some tower coherences already hybridize with the bulk.

Checks: at $\gamma = 0$ the dense spectrum equals $-i(E_m - E_n)$ to 5×10^{-14} ($L = 4, 5$); $\sum \lambda = \mathrm{Tr}\,\mathcal{L} = -\gamma L 4^L$; $L + 1$ eigenstates of H in the tower at energies $h(L/2 - n)$.

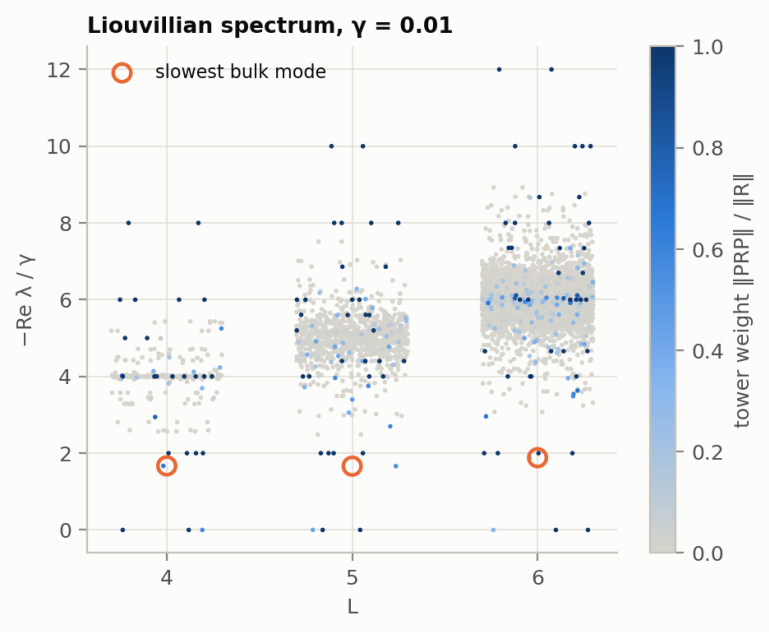


Figure 5: $-\mathrm{Re}\,\lambda/\gamma$ of all Liouvillian modes, coloured by tower weight; circle: slowest bulk mode (`plot_scripts/plot_spectrum.py`).

6 Dependence on γ at fixed $L = 6$

Eleven values $\gamma = 0.002, \ldots, 6$, rates fitted on a window fixed in $\gamma t \in [0.05, 0.6]$ (the window $[5, 60]$ at $\gamma = 0.01$). A window fixed in t samples different parts of the multi-exponential decay at different γ and is not used here.

γ	0.002	0.005	0.01	0.02	0.05	0.1	0.2	0.5	1	3	6
P1 scar	4.80	4.78	4.69	4.61	4.59	5.67	3.75	2.47	2.14	2.02	2.00
P2 energy	2.44	2.42	2.39	2.36	2.26	2.09	1.93	1.62	1.42	1.29	1.27
P3 $\Delta\Gamma$	1.60	1.60	1.60	1.60	1.60	1.61	1.62	1.66	1.72	1.88	2.28

Table 3: Γ/γ at $L = 6$ (random c).

Three regimes for the scar (Fig. 6):

- **weak dephasing**, $\gamma \leq 0.05$: $\Gamma \propto \gamma^{0.984 \pm 0.002}$, i.e. linear, at $\approx 4.7\gamma$;
- **crossover** around $\gamma \approx 0.1$, where $\gamma L \approx h$: a maximum 5.7γ;
- **strong dephasing**, $\gamma \gtrsim 1$: $\Gamma \to 2\gamma$; at $\gamma = 6$, $m_\perp(t) = e^{-2\gamma t}$ to 0.2%. The dephasing is much faster than H_I, so every spin dephases independently.

The non-scar rates are also linear in γ for weak dephasing ($\gamma^{0.977}$ for P2, $\gamma^{1.001}$ for P3). At strong dephasing the energy probe slows down (1.27γ at $\gamma = 6$), presumably because fast dephasing suppresses the coherent dynamics that moves weight out of the protected S^z-like content.

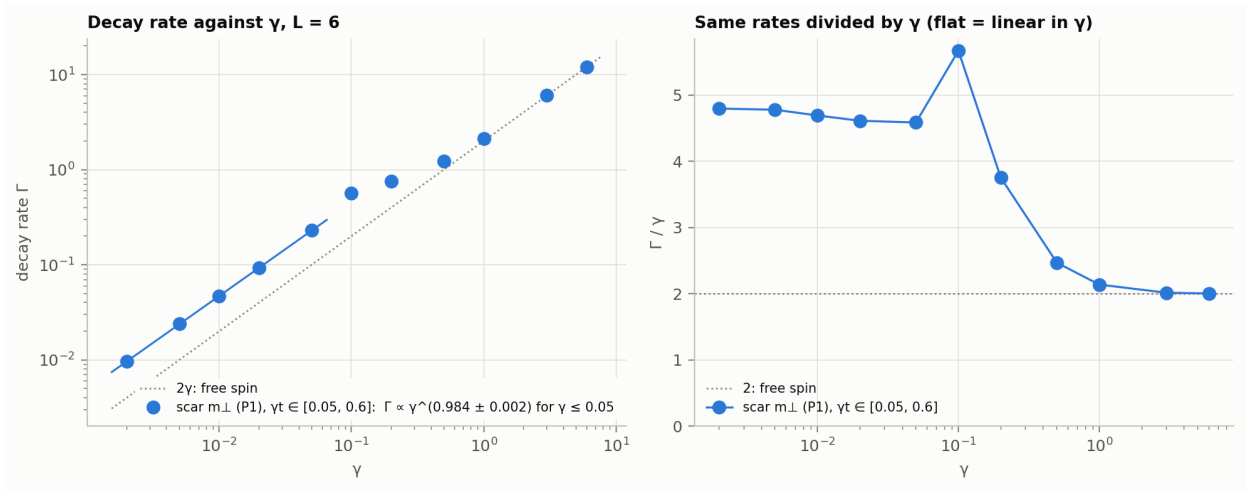


Figure 6: Scar decay rate against γ at $L = 6$; the power law is fitted for $\gamma \leq 0.05$ (`plot_scripts/plot_gamma_scan.py`).

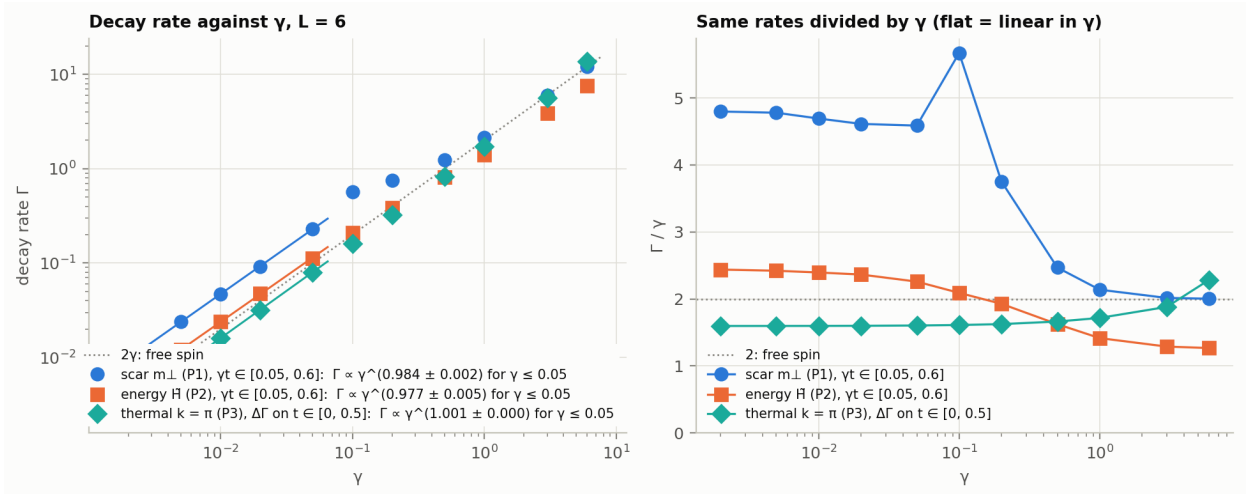


Figure 7: As Fig. 6, with the non-scar probes.

7 Summary and open points

- At $L \leq 8$ and weak dephasing, the scar oscillations decay at a rate growing roughly linearly with L and linearly with γ; non-scar probes decay at $\approx 2\gamma$, nearly independent of L. The L-growth is specific to the scar, consistent with the mid-tower coherences decaying at $\sim \gamma L$.
- This growth is an early-time statement: the exact edge modes decay at 2γ for all L, and the late-time scar decay drifts towards them. Whether the local revival envelope has an L-independent rate in the $L \to \infty$-first limit is not decided by $L \leq 8$.
- The steady state is three-fold degenerate (dark fully polarized states); the project notes assumed uniqueness.
- Open: more disorder realizations; larger L with trajectories; a fit-free observable (e.g. the weights of the exact eigenmodes in $m_\perp$) instead of window-dependent rates; the order of limits $\gamma \to 0$, $L \to \infty$.